University of Dhaka

Department of Computer Science and Engineering

CSE2213: Data and Telecommunication

Design and Implementation of
Amplitude Modulator and Demodulator
on PCB

Lab Group: 03

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1 Introduction

Amplitude Modulation (AM) is a technique in which the amplitude of a high-frequency carrier wave is varied in proportion to the instantaneous amplitude of a lower-frequency information signal. In this project, we designed and built an AM modulator and a corresponding demodulator circuit using simple components, including diodes, inductors, capacitors, and resistors.

The project has two primary objectives:

1. To construct an AM modulator on a printed circuit board (PCB) that takes a carrier signal and a modulating signal as inputs.
2. To construct an AM demodulator on a PCB that recovers the original modulating signal from the modulated carrier.

A functional demonstration of correct modulation on the oscilloscope (and a neat PCB build) are the major evaluation criteria.

2 Circuit Theory and Design

2.1 Modulator

2.1.1 Theory

Amplitude modulation (AM) is a process in which the amplitude of a carrier wave is varied in proportion to the instantaneous amplitude of a modulating signal. This process enables the transmission of information over long distances by embedding the desired signal into a high-frequency carrier wave. A modulator circuit typically achieves this through nonlinear devices and resonant components, which shape the carrier according to the modulating signal. The primary goal is to create a modulated waveform where the amplitude of the carrier varies directly with the modulating signal, while maintaining a stable carrier frequency.

Mathematically, an amplitude-modulated signal can be expressed as:

$$s(t) = [A_c + A_m \cos(\omega_m t)] \cos(\omega_c t)$$

Where:

- A_c is the amplitude of the carrier signal.
- A_m is the amplitude of the modulating signal.
- ω_c is the angular frequency of the carrier.
- ω_m is the angular frequency of the modulating signal.

The process involves key components such as mixing networks, nonlinear devices (e.g., diodes), and tuned circuits to achieve amplitude modulation.

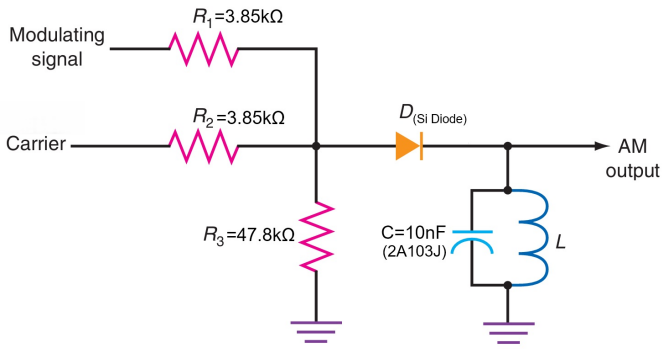


Figure 1: Diode Modulator with an LC Tuned Circuit.

2.1.2 Implementation

A modulator can be realized with a resistive mixing network, a diode, and an LC tuned circuit at the carrier frequency. Figure 1 shows the essential components and connections:

- **Resistive Mixing:** The carrier and the modulating signal are linearly added at the junction of two resistors. While this sum of signals is not truly amplitude modulation (AM is a multiplication process), it sets up the input for the diode stage.
- **Diode Rectification:** The diode is forward-biased only on the positive portions of the composite signal, passing pulses whose amplitudes vary according to the modulating signal.
- **LC Tank Circuit:** Tuned to the carrier frequency. After each diode pulse, the inductor and capacitor exchange energy, effectively "ringing" at the carrier frequency and reconstructing the negative half-cycle. High-amplitude pulses create high-amplitude oscillations, and lower pulses create lower oscillations. The net effect is a carrier whose amplitude is now proportional to the modulating signal—an AM waveform.
- **Q Factor Considerations:** The LC circuit's quality factor must be high enough to filter out unwanted harmonics and produce a clean sinusoidal carrier, yet wide enough to accommodate the sidebands generated by the modulation.

2.1.3 Component Selection

- **Diode:** 1N4148 (fast switching diode).
- **Inductor (L):** Got a shielded inductor that was available on the market and would resonate with the frequencies available at our function generators.
- **Capacitor (C):** 10nF Capacitor paired with L to form an LC resonator.
- **Resistors:** Various values for biasing and limiting current.

2.2 Demodulator

2.2.1 Theory

Demodulation is the process of extracting the original modulating signal from the modulated carrier wave. For this purpose, we employed a classic envelope detector, which is a simple yet effective circuit for amplitude modulation (AM) signal demodulation. The key components of the envelope detector include a diode for rectification and an RC low-pass filter for smoothing the signal.
We chose the time constant $\tau = RC$ such that:

$$\tau \ll \frac{1}{f_{\text{carrier}}} \quad \text{and} \quad \tau \gg \frac{1}{f_{\text{modulating}}}.$$

This ensures the circuit correctly tracks the envelope of the carrier without introducing excessive ripple or distortion.

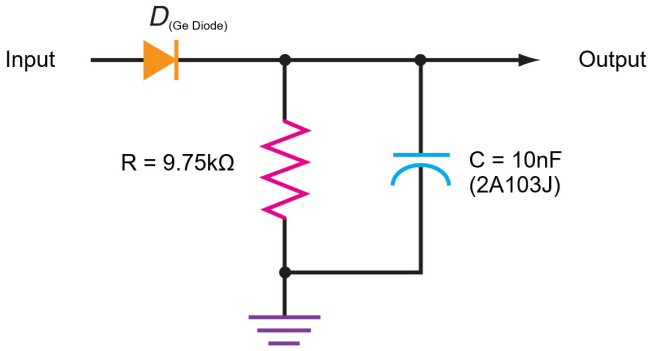


Figure 2: Basic Diode Detector (Envelope Detector).

2.2.2 Implementation

For demodulation, we used a classic envelope detector. Figure 2 shows the essential components and connections.

- **Diode Rectification:** Conducts on positive half-cycles of the AM signal, producing a train of pulses whose peak follows the modulated envelope.
- **RC Low-Pass Filtering:** A capacitor C_1 charges to the peak of each incoming pulse. Between pulses, it discharges slowly through resistor R_1. The time constant $\tau = R_1C_1$ is selected so that the capacitor voltage follows the slower-changing envelope rather than the carrier.
- **Recovered Signal:** The capacitor's voltage is a smooth waveform approximating the original modulating signal. Any high-frequency ripple corresponds to the carrier and is usually negligible if the carrier is significantly higher in frequency than the modulating signal.

2.2.3 Component Selection

- **Diode:** 1N34 (Germanium diode).
- **Capacitor (C):** 10nF Capacitor
- **Resistor (R):** $10k\Omega$ resistor paired with C to create an RC circuit

3 PCB Design and Fabrication

3.1 Layout Considerations

- **Short Traces:** High-frequency signals (carrier) are sensitive to stray inductances; we kept the traces short.
- **Ground Plane:** A solid ground plane on one side helped reduce noise coupling.
- **Component Placement:** Grouped the modulator components (diode, inductor, capacitor) close together. Similarly, the demodulator components (diode, RC filter) are placed in a separate section.

3.2 Etching Process

We used a copper-clad board and the following steps:

1. **Cleansing**: Removed oxidation using fine sandpaper and isopropyl alcohol.
2. **Transfer Method**: We wrote the PCB layout on the cladboard using Nail Polish and as it is resistant to Ferric Chloride, the exposed copper was etched away.
3. **Etching**: Submerged the board in ferric chloride solution, gently agitating until unwanted copper was dissolved.
4. **Drilling and Soldering**: Drilled holes with a 0.9 mm bit, then soldered components carefully, verifying polarity of diodes and capacitors.

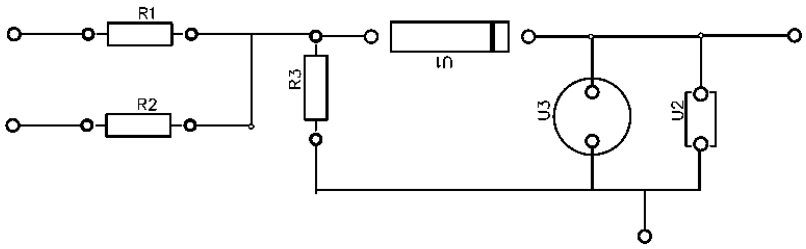


Figure 3: PCB Layout for the AM Modulator

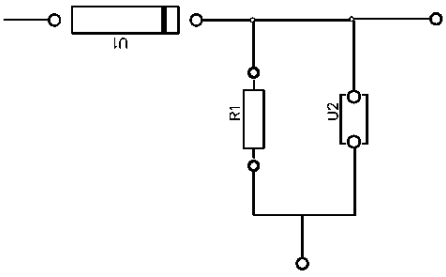


Figure 4: PCB Layout for the AM Demodulator

4 Experimental Setup and Results

4.1 Test Procedure

1. Connected the carrier input (from a function generator at $\sim$ 160kHz) to the modulator input.
2. Provided a low-frequency signal (1 kHz) as the modulating signal.
3. Observed the modulated output on an oscilloscope at the modulator's output node.
4. Fed the modulated signal into the demodulator.
5. Measured the demodulated output on a separate channel of the oscilloscope.

4.2 Oscilloscope Observations

- **Modulator Output:** Showed a carrier wave whose amplitude varied in proportion to the audio input.
- **Demodulator Output:** After the diode detector, the recovered envelope totally matched the original modulating signal.

4.3 Demonstration Video

Drive link to the working demonstration of the project

4.4 Performance Analysis

- The diode-based modulator successfully varied the carrier amplitude in accordance with the input signal.
- The demodulator recovered the modulating signal with minimal distortion. Some fine-tuning of the RC time constant was done to optimize the fidelity.
- PCB construction was stable and reduced noise compared to a breadboard prototype.

5 Discussion

5.1 Modulator Performance

The diode modulator effectively performed the multiplication-like operation via:

- Linear addition of carrier and modulating signals,
- Diode half-wave rectification,
- LC "ringing" to reconstruct the negative half cycle.

Ensuring the LC tank circuit has a suitable Q factor was crucial to achieve a clean carrier without excessive harmonics.

5.2 Demodulator Performance

The envelope detector successfully recovered the audio signal:

- The time constant R_1C_1 was chosen to be long relative to the carrier period and short relative to the modulating period.
- Only minor ripple was observed, consistent with typical envelope detection theory.

5.3 Challenges

- **LC Tuning:** Finding the correct carrier frequency for the available inductor-capacitor pair required iterative testing and prior calculation.
- **Component Tolerances:** Real inductors and capacitors have tolerances that slightly shift the resonant frequency.
- **PCB Etching Quality:** Ensuring no shorts or incomplete etches was critical for a clean, noise-free circuit.

5.4 Possible Improvements

- **Buffer Amplification:** Inserting a buffer amplifier stage before or after the LC tank could improve signal strength.
- **Automatic Gain Control (AGC):** To stabilize amplitude variations for real-world applications.

6 Conclusion

In this lab project, we successfully designed, fabricated, and tested a diode-based AM modulator with an LC circuit and a corresponding diode-based envelope detector. The results on the oscilloscope confirm that the amplitude modulation and subsequent demodulation operate correctly, demonstrating the key principles of AM transmission. This hands-on experience solidified our understanding of modulation theory, circuit design, and PCB fabrication techniques.

7 References

- B. A. Forouzan, *Data Communications and Networking*, 4th ed., New York: McGraw-Hill, 2007.
- L. E. Frenzel, *Principles of Electronic Communication Systems*, 3rd ed., New York: McGraw-Hill, 2012, ch. 4.